

MicroPicoDrive NG hardware installation manual

This manual covers fitting the MicroPicoDrive NG mainboard inside a Sinclair QL, including the Pico and the optional vibration motor. Day-to-day use, SD card preparation, System Tools and firmware updates are covered by the firmware manual in the [firmware repository](#).

How complete is this guide? The operating steps have been checked against the firmware, and the photographs show the current NG boards, the Pico orientation and the optional motor module. What we could not yet confirm is the QL ribbon orientation, the mounting hardware and the exact fitting steps for each drive bay — so treat this guide as an outline, and follow the instructions supplied with your board revision for those details.

The two parts and firmware versions

The **mainboard** stays inside the QL and connects to its Microdrive bus. It carries the Raspberry Pi Pico that runs the firmware. The removable **cartridge board** carries the display, four buttons and microSD socket. It plugs into the mainboard's edge connector.

A **cartridge image** is a file on the microSD card that behaves like one QL Microdrive cartridge. When you *mount* an image, the QL sees it as a cartridge sitting in the drive; when you *eject* it, you are back in the file browser. Mounting and ejecting happen on the screen — they are not the same thing as physically plugging in or pulling out the cartridge board.

Feature	Lite: Raspberry Pi Pico / RP2040	Full: Raspberry Pi Pico 2 W / RP2350
Image browsing, QL loading and saving	Yes	Yes
Display and motor settings	Yes	Yes
SD card firmware updates	UF2 file	BIN file and matching JSON manifest
Bluetooth file management and updates	No	Yes
Revert Firmware	No	When another firmware slot is available
BOOTSEL Mode menu entry	Yes	No

Install the firmware that matches the Pico you actually fitted. The Pico 2 W's wireless features use Bluetooth Low Energy (BLE), not Wi-Fi.

Installing the mainboard in the QL

Before opening the case

What you'll need before you open the case:

- The assembled mainboard and cartridge board
- The correct QL ribbon cable and mounting hardware for your kit
- A prepared microSD card

If your Pico came blank, plan to install the firmware before you close the QL up again — see the **Firmware updates** section of the firmware manual.

1. Shut down the QL, disconnect its power supply and disconnect attached equipment. Remove any tape cartridges.
2. Work on a clean surface and handle the boards by their edges. Keep loose screws and other metal away from the electronics.
3. Open the QL using the service instructions for its case revision. Lift the cover carefully: the keyboard membrane tails connect it to the motherboard. Avoid pulling, sharply bending or trapping these tails.
4. Record the existing Microdrive cable routing and connector orientation before disconnecting anything.

Fitting outline

The mainboard replaces an internal Microdrive mechanism. Its QL bus connector is **J2**, a 14-pin (2 by 7) header. **J1** is the cartridge edge connector. The optional motor connection is **J3**, the small three-pin socket marked **GND**, **VCC** and **IN**. Confirm these markings against your actual board revision.



Rear of the NG mainboard and cartridge board, showing the Pico, J3 motor socket, edge connector and microSD socket.

Fitting the Pico

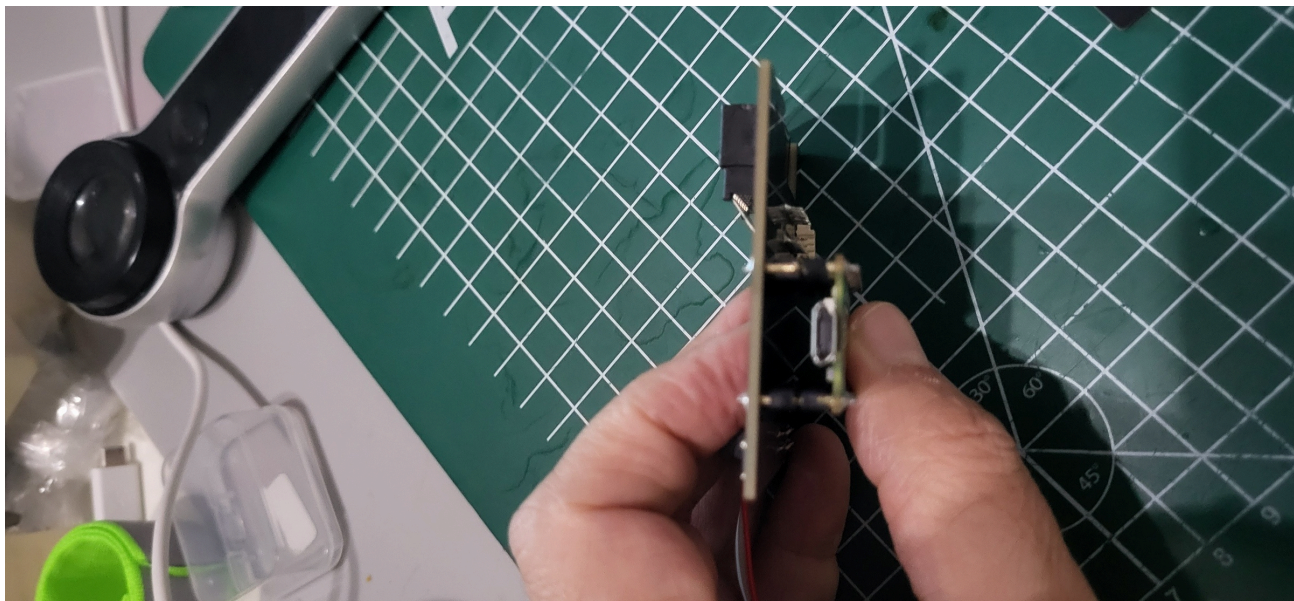
If the mainboard was supplied without a Pico, fit the correct board before installing the assembly in the QL. Use a Raspberry Pi Pico for Lite firmware or a Pico 2 W for Full firmware.

1. Disconnect QL and USB power. Hold the Pico by its edges and avoid touching the contacts.
2. Orient it as shown below: the Pico's USB connector sits at the same end as the mainboard's J3 motor socket. Check the orientation before engaging any pins.
3. Align both rows with the sockets. Start every pin squarely, then press the board down evenly with light pressure at both ends. Stop if a pin bends or the rows do not enter together.
4. Look along both sides to confirm that no pin is outside a socket and that the Pico is fully and evenly seated.



Aligning a Pico 2 W with the two mainboard sockets; its USB connector is beside the J3 motor socket end.

The Pico sits underneath the mainboard. That low profile is what leaves room for the QL keyboard above it, so don't add tall headers or spacers unless your enclosure's fitting instructions say you can.



Side view of the Pico mounted beneath the mainboard, showing the intended low-profile clearance.

Installing and connecting the optional motor

Use a **driver-equipped vibration module** made for a logic-level trigger and a 5 V supply, such as the module shown below. J3 is not a bare-motor output: connecting a motor directly to it can overload GP11 and damage the Pico.

J3 is a three-pin Molex PicoBlade socket. Its electrical connections are:

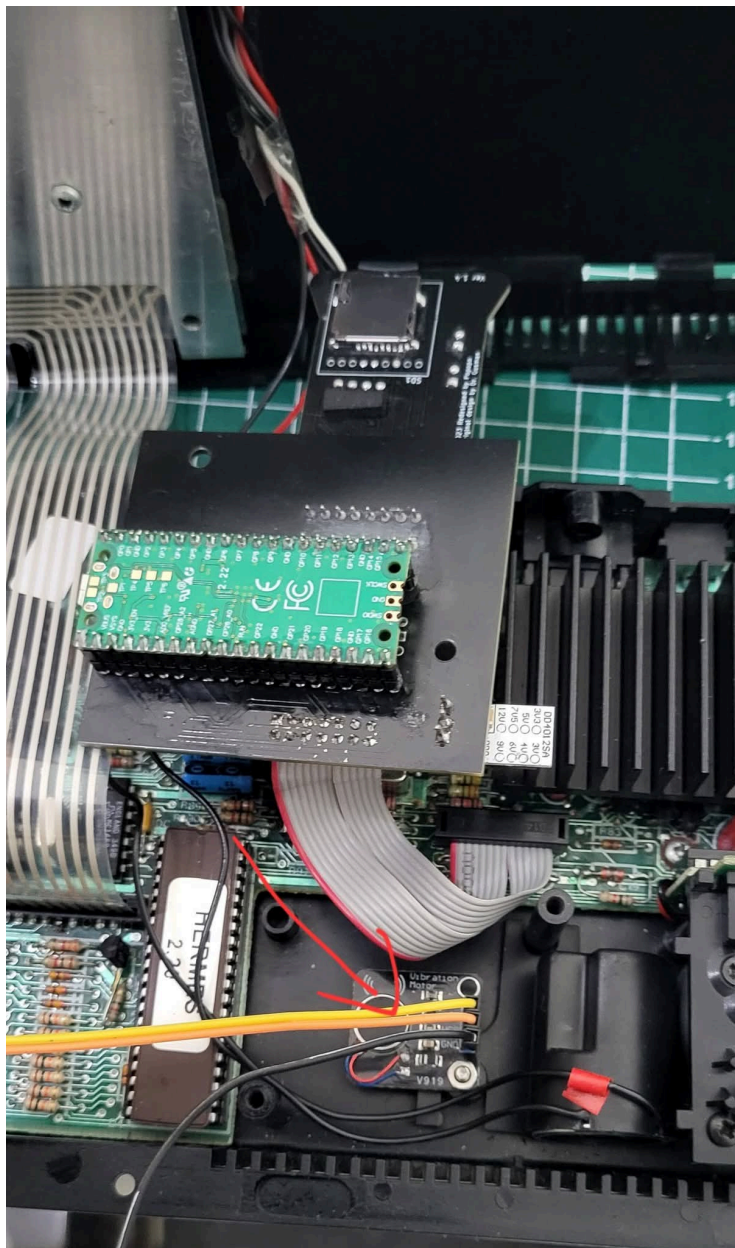
J3 connection	Function
GND	Ground
VCC	+5 V supply for the motor module
IN	Logic input from Pico GP11



Driver-equipped coin vibration motor module and its three-wire GND, VCC and trigger connection.

1. Disconnect QL and power it off before connecting or moving the motor.

2. Inspect the labels at both ends. Connect J3 **GND** to the module's ground, J3 **VCC** to its supply input, and J3 **IN** to its trigger input. Use the supplied keyed lead where available. **Do not rely on wire colour alone**; cable colours and module pin order can differ.
3. Check that the plug is fully seated and that no contact is shifted sideways. Keep the lead away from the cartridge opening, keyboard membrane tails, sharp edges and screw posts.
4. Before fixing the module permanently, close or support the keyboard safely, power the QL and run **System Tools > LED & Motor Test**. The motor should run for five seconds. Switch off and disconnect power again before adjusting it.
5. Removing the original Microdrive exposes a support hole in the lower case. Place the motor module over that support as shown below and align one of its mounting holes with the case support hole. Pass the **supplied screw** through the aligned holes, fit the **supplied nut** on the opposite side, and tighten until the module is secure. Do not overtighten: the module PCB and plastic support can be damaged. Check that the module lies flat and that its solder joints cannot touch the QL motherboard or nearby metalwork.
6. Re-run LED & Motor Test after final assembly. Configure normal feedback under **System Tools > Motor** as described in the firmware manual. Leave enough cable slack for servicing, but keep the lead out of the keyboard, cartridge and case-screw paths.



Motor module secured to the lower-case support hole exposed after removing the original Microdrive, using the supplied screw and nut.

Fitting the mainboard

1. Identify the drive position to be replaced using the kit's fitting instructions. The firmware has no menu setting for choosing `mdv1_` or `mdv2_`; numbering depends on the QL's drive-select chain and wiring.
2. Remove the selected mechanism as directed for the kit. Retain its hardware separately so it can be restored later.
3. Fit the mainboard with the specified supports and fasteners, with its cartridge connector aligned to the case opening. Check that the underside cannot touch conductive parts and that inserting a cartridge will not bend an unsupported board. Do not assume the original screws are the right length.
4. Connect the QL ribbon to J2 only after confirming pin 1 and the cable orientation at **both ends** against the board-specific fitting instructions. Check that neither connector is offset by a row or a pin. The stripe on the cable is not, by itself, proof of the right orientation.
5. If required, fit the Pico and optional motor as described above. Make these connections with power disconnected.
6. Route cables clear of case posts, sharp edges and the cartridge opening. Check connector seating, board clearance and the keyboard connections before refitting the cover.
7. Insert the prepared cartridge board without force, reconnect the QL and perform the first-start check described in the **First start and everyday use** section of the firmware manual.

Only the removable cartridge board is designed for powered insertion and removal. Disconnect QL power before changing the mainboard, Pico, ribbon or motor wiring. USB servicing also needs a board-specific power arrangement; do not assume simultaneous USB and QL power is supported.

Hardware drawings, schematics and board photographs live in this repository's board folders, `MicroPicoDrive/` and `MicroPicoDriveCartridge/`.

If something goes wrong after reassembly

- **No display:** check cartridge-board seating and QL power. Disconnect power

before inspecting internal ribbon or Pico connections.

- **QL cannot read the image:** wait for mounting, use the correct `mdv`

number and try a known-good image; inspect wiring only with power disconnected.

See the firmware manual's Troubleshooting section for everything else.